NGUYEN VAN HUNG

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SUMMARY

I am actively pursuing an AI Engineer position, bringing hands-on experience across three key projects spanning user behavior prediction, server traffic estimation for auto-scaling, and deep learning in signal processing. I drive real impact by leveraging advanced data analysis and model optimization techniques to build high-performance, production-ready AI systems.

EDUCATION

University of Science (VNU-HN) **9/2024 – 8/2028 (Expected)**

Major: Computer Science

Relevant Coursework: Data Structures , Object-Oriented Programming, Machine Learning, Deep Learning, Database Systems, Probability and Statistics.

EXPERIENCE

Institute of Physics, Vietnam Academy of Science and Technology 5/2026 - Present

Collected and processed real-world acoustic signals using digital signal processing techniques to prepare datasets for deep learning model development. Developed and evaluated deep learning models, explored optimization methods to reduce computational complexity, and deployed optimized models on edge computing devices for efficient inference.

PROJECT

Autoscaling Forecasting & Optimization 1/2026 - 4/2026

- Responsibilities: Processed and pre-processed ~2.9 million Apache log requests, engineered 13 advanced features, trained forecasting models (XGBoost, LightGBM, Hybrid LSTM-Prophet), and designed a 4-layer Hybrid Autoscaling system combining multi-instance allocation strategies to optimize cloud resources.
- Achievements: Achieved top performance with the Hybrid model (R^2 = 0.94, MAPE = 7.2%), reduced cloud infrastructure costs by 25–35%, cut resource waste from 60% down to 20%, and maintained SLA compliance above 99.2%.
- Limitations: Relied on historical training data from 1995 which does not fully reflect modern user behavior, lacked automated concept drift detection, and simplified hardware assumptions by excluding non-CPU factors (such as I/O and memory).

Research Acoustic-Based Durian Ripeness Classification System 5/2026 - Present

- Data Collection: Collected acoustic data from 28 durian samples using an ICS-43434 microphone and ESP32-S3, obtaining 1,126 knocking sound samples labeled as ripe or unripe based on reference measurements.
- Data Processing: Applied signal smoothing, peak detection (15% amplitude threshold), and event segmentation. Extracted log-scaled Mel-Spectrogram features (128 × 16) and applied Gaussian noise injection ($\sigma = 0.001–0.01$), time shifting, and brightness adjustment to balance the dataset to 1,700 samples (850 per class), split into 70% training, 15% validation, and 15% testing.
- Achievements: Achieved a test accuracy of 94.90% and an F1-score of 0.95 with the baseline CNN model. Successfully compressed the model size by nearly 50x (reducing parameters from 8.91M to 0.158M, a 98.2% reduction) via Knowledge Distillation and Structured Pruning while maintaining a high accuracy of 94.90%.

ACHIEVEMENTS

- Achieved a Top 10 ranking in the DATAFLOW competition.

SKILLS

- Languages: Python, C/C++
- Machine Learning & Data: PyTorch, Scikit-learn, NumPy, Pandas
- APIs & Web: FastAPI, Flask
- Databases & Storage: SQL, NoSQL
- DevOps & Tools: Docker, Git/GitHub, Linux